Python Activity-Based Learning
Case Study 31 String-Based Examination Answer Evaluation

Selected case study | Prepared using the topics listed in the supplied assignment.

1. Problem Statement

Compare a student's answers with an answer key, count correct, incorrect, and unanswered questions, apply negative marking, and calculate the final score.

2. Objectives

• Store answer strings in lists.

• Compare student answers with an answer key.

• Count correct, incorrect, and blank answers.

• Calculate marks with negative marking.

3. Assumptions

• There are five multiple-choice questions.

• Each correct answer earns 2 marks.

• Each incorrect answer loses 0.5 marks.

• An unanswered question earns 0 marks.

• Answers are A, B, C, or D; blank input is allowed.

4. Algorithm

1. Store the answer key in a list.

2. Read five student answers and convert them to uppercase.

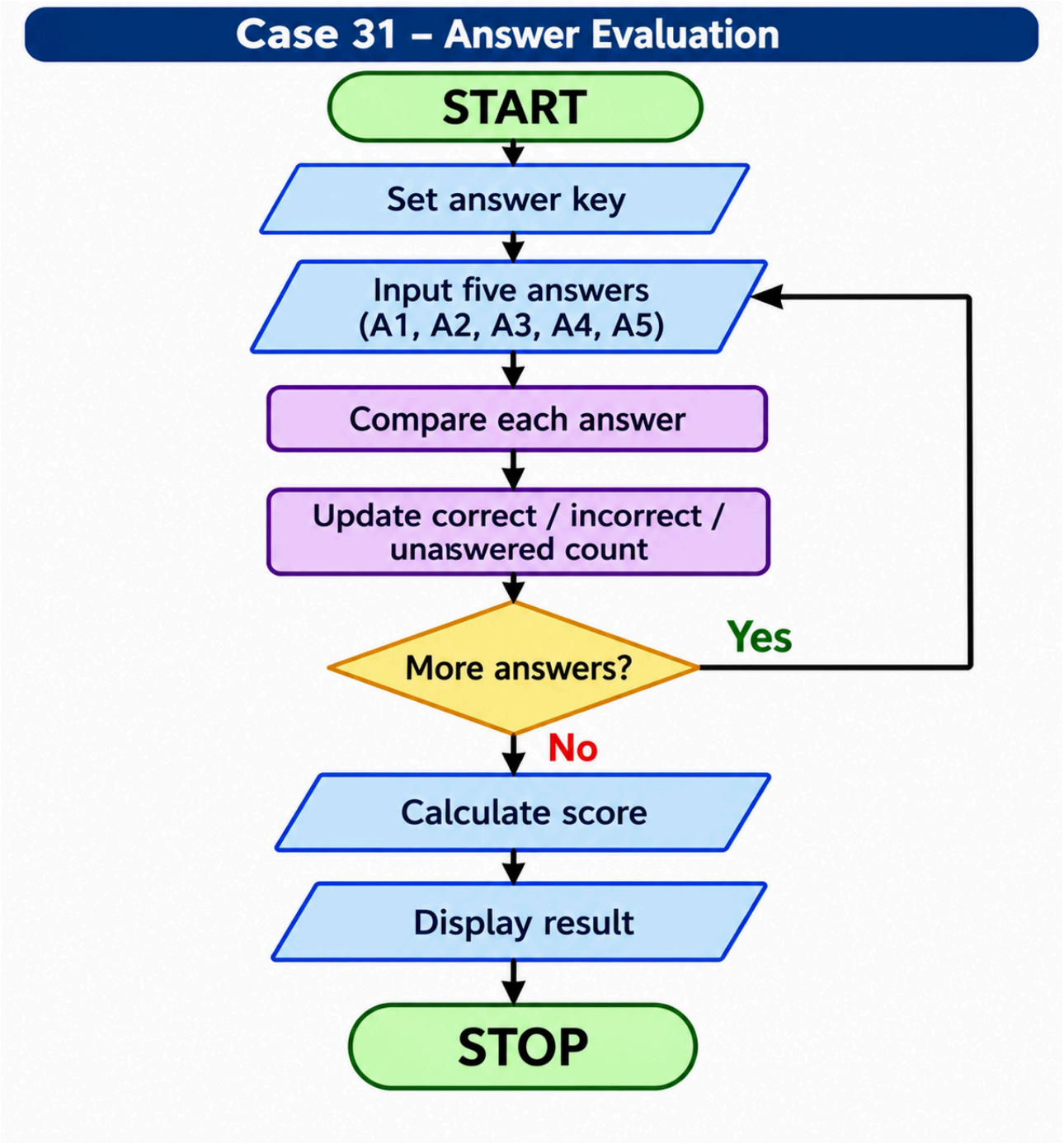
3. Compare each answer with the corresponding key.

4. Update correct, incorrect, or unanswered counters.

5. Calculate score as 2 times correct minus 0.5 times incorrect.

6. Display counts and final score.

5. Flowchart



6. Pseudocode

```
BEGIN
 key ← [A, C, B, D, A]
 INPUT student answers
 correct ← 0; incorrect ← 0; blank ← 0
 FOR each answer position
 IF answer is blank THEN blank++
 ELSE IF answer = key[position] THEN correct++
 ELSE incorrect++
 END IF
 END FOR
 score ← correct × 2 - incorrect × 0.5
 PRINT counts and score
```

```
END
```

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7. Python Source Code

```
answer_key = ["A", "C", "B", "D", "A"]
student_answers = []

for i in range(5):
 answer = input("Answer for question " + str(i + 1) + " (A/B/C/D or blank): ").strip().upper()
student_answers.append(answer)

correct = 0
incorrect = 0
unanswered = 0

for i in range(5):
 if student_answers[i] == "":
 unanswered += 1
 elif student_answers[i] == answer_key[i]:
 correct += 1
 else:
 incorrect += 1

score = correct * 2 - incorrect * 0.5

print("\nCorrect answers:", correct)
print("Incorrect answers:", incorrect)
print("Unanswered:", unanswered)
print("Final score:", score, "out of 10")
```

8. Test Cases

Input / Scenario	Expected result
A,C,B,D,A	5 correct; score 10
A,B,B,D,A	4 correct; 1 incorrect; score 7.5
blank,blank,blank,blank,blank	0 correct; 0 incorrect; 5 unanswered; score 0

9. Conclusion

The program uses strings, lists, loops, comparisons, and arithmetic to evaluate an objective examination.

Note: Run the program in Python 3. The sample test cases are illustrative; enter the values when

prompted.